Computer Vision For Volleyball Serve Analytics

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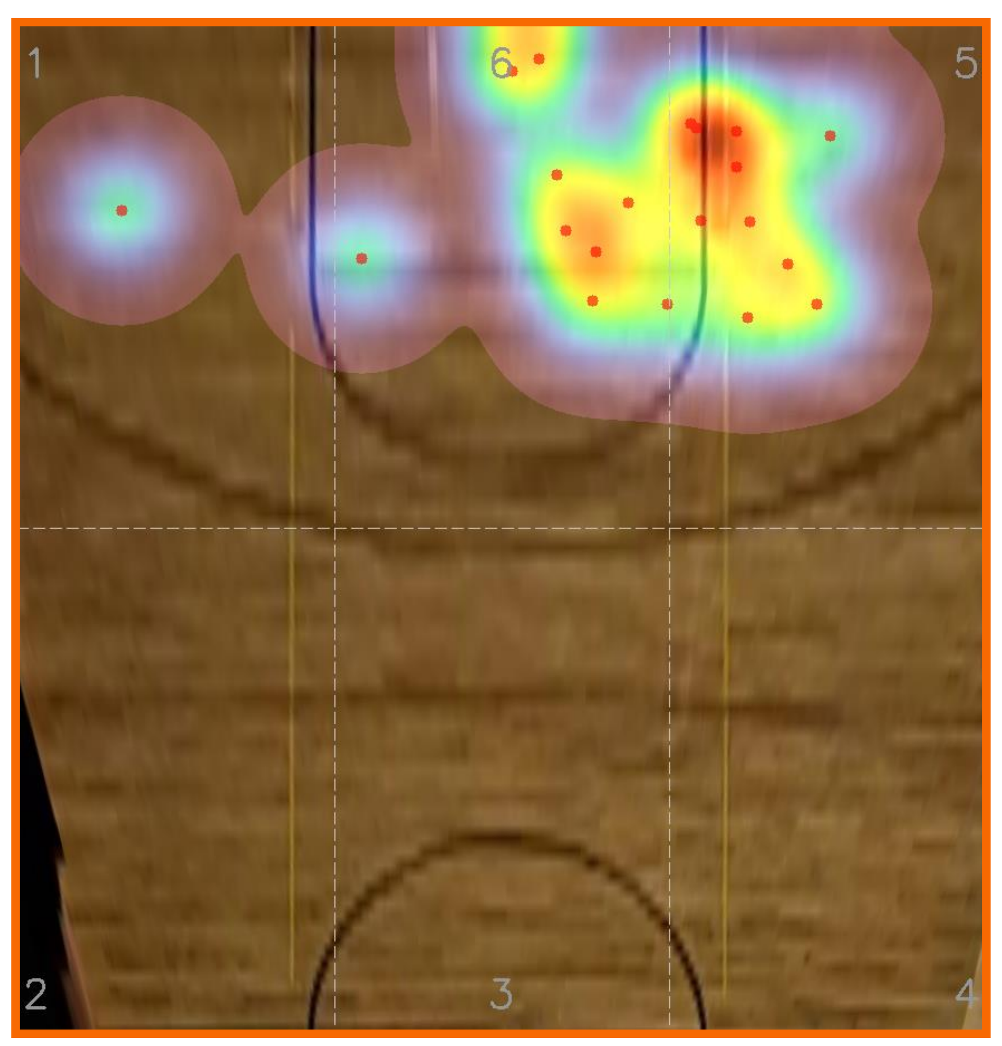
Introduction

This project develops a **single-camera system** for automated volleyball serve analysis. The tool processes standard baseline video and extracts **objective** measurements of a serve's flight path, landing location, and overall consistency.

Traditional review is slow and subjective, players scrub through footage and estimate landing spots by eye, while this system provides **precise, repeatable metrics** from ordinary practice recordings.

The system performs:

- Ball detection and temporal tracking
- Full trajectory reconstruction
- Hit-frame and landing-frame estimation
- Court-mapped landing coordinates
- Automated statistics and visualization



Data Collection

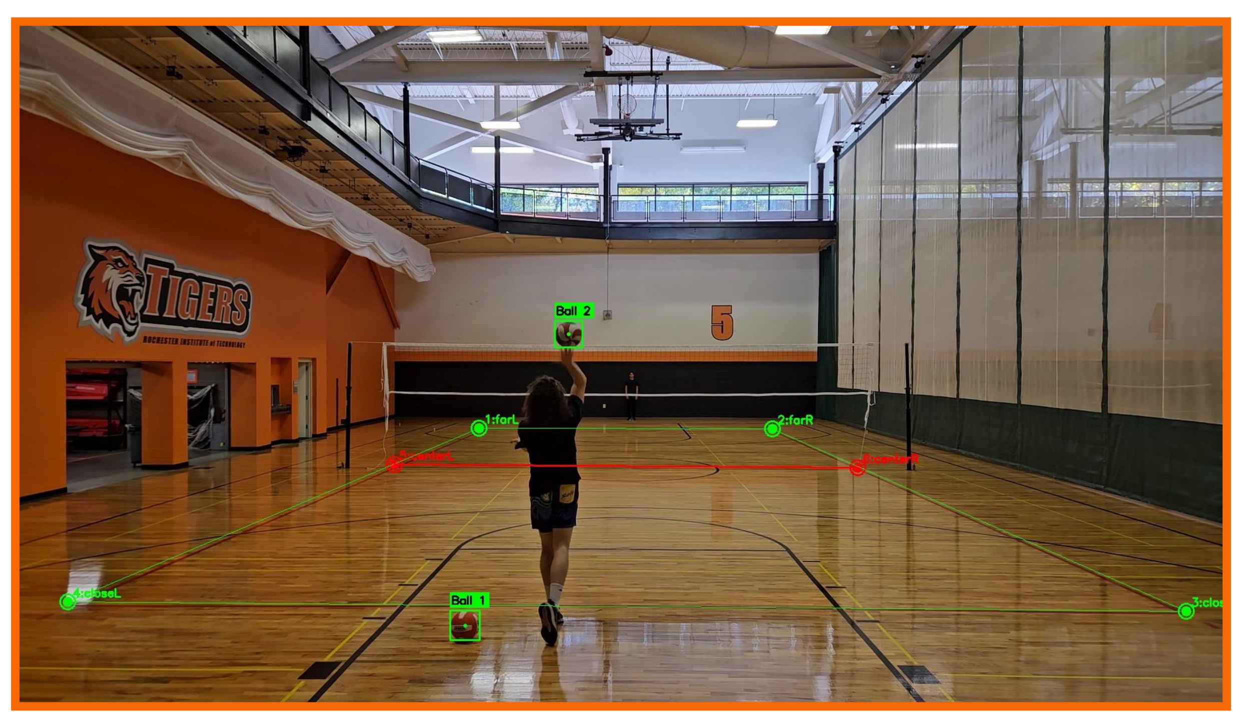
Data was collected using a smartphone camera positioned at the center of the baseline, 4 meters behind the court and 74 inches high. All sessions were recorded at 1080p, 60 FPS, ensuring the entire court was in frame.

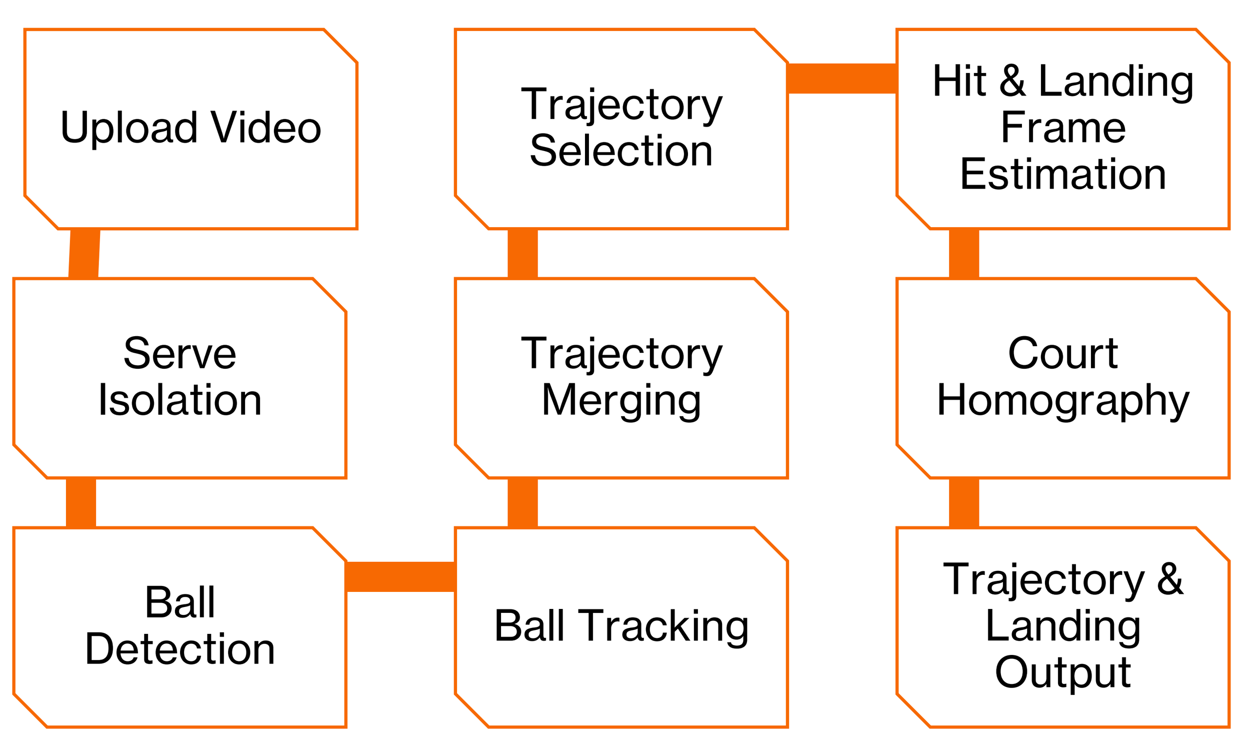
Annotations include:

- Six court-corner points per session
- Ball bounding boxes per frame

The dataset consists of:

- 5 recordings
- 5 ball types
- 113 serves
- 16,172 frames

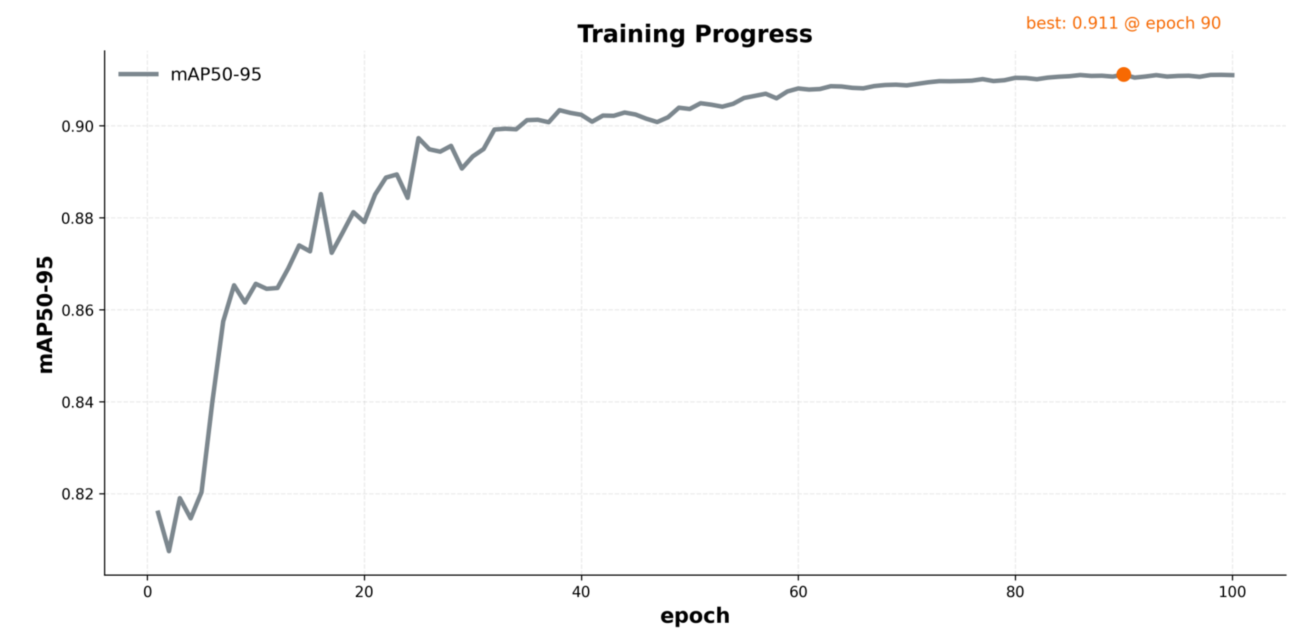




Pipeline

1) **Serve Isolation**: Footage is split into individual serve clips.
2) **Ball Detection**: A custom-trained YOLOv8 model detects the ball in every frame.
3) **Ball Tracking**: SORT links detections across time and produces short trajectory fragments.
4) **Trajectory Merging**: Fragments are combined using a greedy algorithm that minimizes gaps.
5) **Final Trajectory Selection**: The merged chain with stable motion and realistic size decay is chosen as the serve.
6) **Hit & Landing Frame Estimation**: Ball motion and size are smoothed with a Savitzky–Golay filter to estimate the hit and landing frames.
7) **Court Homography**: Using six user-labeled court corners, pixel coordinates are projected into real space.
8) **Trajectory & Landing Output**: The system generates plots, heatmaps, and serve-level statistics for interpretation.

Evaluation



F1	Precision	Recall
99.489	99.452	99.527

Landing-frame error is typically within 1 frame.

Conclusion

This project demonstrates that accurate volleyball serve analysis can be achieved using a single baseline camera. **YOLOv8** and **SORT** provide reliable ball tracking, while **homography** enables real-court trajectory and landing estimation. The system offers accessible, objective data that helps players evaluate accuracy, consistency, and improvement over time.

Future Work

- Automatic serve isolation using MediaPipe Pose estimation
- Automatic court line detection using U-Net
- Session-over-session improvement tracking